

# Pushing the Planar Graph-Reduction Step

May 2026

## 1 Executive summary

This note tries to prove the remaining planar reduction-to-graph step. I do not get a complete proof. I do get a sharper formulation that looks more realistic than the previous  $O(\delta_T)$ -radialization target.

The key adjustment is to stop at a boundary layer of size

$$\lambda \simeq \sqrt{\delta_T(\Omega)}$$

rather than at size  $O(\delta_T(\Omega))$ . This is still sharp for the final asymmetry estimate, because the transfer back costs  $O(\lambda)$  in Fraenkel asymmetry and hence  $O(\lambda^2) = O(\delta_T)$  after squaring. The gain is that the level-set Chebyshev step now gives

$$D_H(\hat{t}) + D_I(\hat{t}) \lesssim \sqrt{\delta_T(\Omega)}$$

instead of only a fixed small constant. In dimension two this gives real geometric information: small components have total area  $O(\delta_T)$ , holes with small perimeter have total area  $O(\delta_T)$ , and connected outward fingers of macroscopic length are excluded by perimeter unless they are exactly the thin objects that the  $D_H$  weight detects.

The proof still has one missing arbitrary-geometry lemma: a *good-level convexification/radialization lemma*. It must say that convexifying, filling, or pruning only the non-graph pieces of the good level changes the normalized torsion deficit by  $O(\delta_T)$ . The model estimates support this, but I do not see a complete decomposition proof in the current repo.

## 2 The better stopping scale

Let  $|\Omega| = \pi$  and set  $\delta = \delta_T(\Omega)$ . The volume-variable identity from Plan 3 Step 1 gives, for  $s = m(t)$ ,

$$\int_0^\pi \frac{D_H(v(s)) + D_I(v(s))}{c s^0} ds \leq C\delta \quad (n = 2).$$

Therefore for every layer budget  $\lambda \in (0, \pi/4)$  there is a regular level  $E_t$  with

$$|\Omega \setminus E_t| \leq \lambda, \quad D_H(t) + D_I(t) \leq C \frac{\delta}{\lambda}. \quad (2.1)$$

The Talenti rearrangement bound also gives

$$t \leq C\lambda. \quad (2.2)$$

Indeed, the ball level with removed area  $\lambda$  has height  $\lambda/(4\pi) + O(\lambda^2)$ , and  $u_\Omega^* \leq u_B^*$ .

Taking

$$\lambda = \kappa\sqrt{\delta}$$

gives a regular level  $E = E_{\hat{t}}$  satisfying

$$|\Omega \setminus E| \leq C\sqrt{\delta}, \quad D_H(E) + D_I(E) \leq C\sqrt{\delta}, \quad \delta_T(E) \leq C\delta. \quad (2.3)$$

The layer transfer back to  $\Omega$  is still sharp:

$$\mathcal{A}(\Omega) \leq \mathcal{A}(E) + C\sqrt{\delta}, \quad \mathcal{A}(\Omega)^2 \leq 2\mathcal{A}(E)^2 + C\delta. \quad (2.4)$$

Thus it is enough to prove  $\mathcal{A}(E)^2 \leq C\delta$  for the  $\sqrt{\delta}$ -stopped level.

**Remark 2.1.** *This is a genuine improvement over the previous  $O(\delta)$ -layer graph entry target. At an  $O(\delta)$  layer, Chebyshev only gives a fixed small  $D_H + D_I$ . At a  $\sqrt{\delta}$  layer, the geometric defects tend to zero. The final exponent is not harmed because asymmetry, not volume, is squared at the end.*

### 3 What $D_I$ already proves in the plane

Write

$$\varepsilon := C\sqrt{\delta}$$

for the right side of  $D_I(E) \leq \varepsilon$ . Since

$$D_I(E) = P(E)^2 - 4\pi|E|,$$

and  $|E| = \pi + O(\sqrt{\delta})$ , after harmless rescaling we may read this as

$$P(E) \leq 2\pi + C\varepsilon. \quad (3.1)$$

**Lemma 3.1** (Small components are negligible). *Let  $E \subset \mathbb{R}^2$  have  $|E| = \pi$  and  $P(E) \leq 2\pi + \varepsilon$ , with  $\varepsilon \leq \varepsilon_0$ . Let  $E_0$  be an indecomposable component of largest area and set  $E_{\text{sat}} = E \setminus E_0$ . Then*

$$|E_{\text{sat}}| \leq C\varepsilon^2.$$

*Proof.* Let  $a_j$  be the areas of the indecomposable components, ordered so that  $a_0$  is largest, and let  $v = \sum_{j \geq 1} a_j$ . By the planar isoperimetric inequality and additivity of perimeter over indecomposable components,

$$2\pi + \varepsilon \geq P(E) \geq 2\sqrt{\pi} \left( \sqrt{\pi - v} + \sum_{j \geq 1} \sqrt{a_j} \right).$$

For small  $v$ ,

$$2\sqrt{\pi}\sqrt{\pi - v} \geq 2\pi - Cv.$$

Hence

$$\sum_{j \geq 1} \sqrt{a_j} \leq C\varepsilon + Cv.$$

For  $\varepsilon_0$  small,  $v \leq \pi$  and the  $Cv$  term is absorbed using  $v \leq (\sum_{j \geq 1} \sqrt{a_j})^2$  unless  $v$  is already bounded by  $C\varepsilon^2$ . Thus  $v \leq C\varepsilon^2$ .  $\square$

With  $\varepsilon \simeq \sqrt{\delta}$ , disconnected satellites have total area  $O(\delta)$ . They can be deleted without affecting the final asymmetry bound. Their torsional rigidity is also negligible:

$$T(E_{\text{sat}}) \leq \sum_j \frac{|E_j|^2}{4\pi} \leq \frac{|E_{\text{sat}}|^2}{4\pi} = O(\delta^2),$$

by Saint–Venant on each component.

**Lemma 3.2** (Small holes have quadratic area). *Let  $F$  be the main component and suppose the total perimeter of bounded hole boundaries is at most  $p_h$ . Then the total area of those holes is*

$$|H_{\text{holes}}| \leq \frac{p_h^2}{4\pi}.$$

*Proof.* Apply the planar isoperimetric inequality to each hole  $H_j$ :  $|H_j| \leq P(H_j)^2/(4\pi)$ . Summing and using  $\sum_j P(H_j)^2 \leq (\sum_j P(H_j))^2$  gives the claim.  $\square$

Thus closed holes whose total boundary cost is  $O(\sqrt{\delta})$  have total area  $O(\delta)$ . This is already enough for tiny interior holes such as a punctured disk. The capacity mechanism from `PLANAR_RADIALIZATION_PROOF_ATTEMPT.tex` is stronger, but for the  $\sqrt{\delta}$ -stopped level the perimeter estimate already gives the correct area scale for closed holes.

## 4 Convex hull and the graph envelope

Let  $F$  be the main connected component from Lemma 3.1, and let  $K = \text{co}(F)$ . For planar continua,

$$P(K) \leq P(F). \tag{4.1}$$

Consequently  $K$  is a convex set with perimeter  $2\pi + O(\varepsilon)$  and area  $\pi + O(\varepsilon)$ . By Bonnesen’s inequality, after translating to a suitable center  $z$ , its inradius and circumradius satisfy

$$R_K - r_K \leq C\sqrt{\varepsilon}. \tag{4.2}$$

For  $\delta$  small this puts  $K$ , after area normalization, in the small-amplitude radial-graph class:

$$K^\sharp = \{z + re^{i\theta} : 0 < r < 1 + \phi_K(\theta)\}, \quad \|\phi_K\|_{L^\infty} \leq C\sqrt{\varepsilon}. \tag{4.3}$$

This is a useful reduction but not yet a proof. The convex hull can overfill sparse outward structures. If  $K \setminus F$  has volume merely  $O(\varepsilon) = O(\sqrt{\delta})$ , the crude fill-rescale lemma gives only

$$\delta_T(K^\sharp) \leq C(\delta_T(F) + \sqrt{\delta}),$$

which is too weak. To close the sharp result one needs the stronger statement

$$\delta_T(K^\sharp) \leq C\delta_T(E). \tag{4.4}$$

This is exactly where the remaining graph-reduction work sits.

## 5 Why $D_H$ is the right extra input

The obstruction to using the convex hull alone is a sparse outward finger. In the model case, take a tube of width  $w$  and length  $L$  attached to the disk. The convex hull may add area comparable to  $L$  times the angular span, while the actual tube volume is only  $wL$ . Perimeter sees only  $L$ , so choosing the stopped level with  $D_I = O(\sqrt{\delta})$  permits fingers of length  $L = O(\sqrt{\delta})$ , but not longer. This already makes the hull a small-amplitude graph.

The sharp deficit comparison, however, uses the  $D_H$  weight. On active levels inside a planar tube,

$$D_H \simeq \frac{L}{w}. \quad (5.1)$$

Thus a long thin tube cannot occur on a  $D_H$ -good level. Equivalently, if the tube survives the level  $u = t$ , then either:

1. its cross-width satisfies  $w^2 \gtrsim t$ , since the torsion height in a width- $w$  tube is  $O(w^2)$ ; or
2. it lies below the stopped level and has already been removed.

At the  $\sqrt{\delta}$ -stopped level,  $t \leq C\sqrt{\delta}$ . Therefore any surviving tube has  $w \gtrsim \delta^{1/4}$ . Since the torsion deficit of a planar outward tube is  $O(wL)$ , the length of a surviving tube paid by  $\delta$  is at most

$$L \lesssim \frac{\delta}{w} \lesssim \delta^{3/4}.$$

So the dangerous long-thin convexification example is not compatible with the good-level information.

This model computation is the reason the reduction should be true in dimension two. It also shows why the proof must use both pieces:

$$D_I \quad \text{for perimeter/topological cleanup,} \quad D_H \quad \text{for capillary thin-finger cleanup.}$$

## 6 A sufficient missing lemma

The following lemma would close the planar graph reduction without using BDV.

**Conjecture 6.1** (Good-level convexification lemma). *Let  $E = E_{\hat{t}}$  be a regular  $\sqrt{\delta}$ -stopped good level in dimension two, satisfying*

$$|\Omega \setminus E| \leq C\sqrt{\delta}, \quad D_H(E) + D_I(E) \leq C\sqrt{\delta}, \quad \delta_T(E) \leq C\delta.$$

*Let  $F$  be its largest indecomposable component and  $K = \text{co}(F)$ . After deleting the satellite components of total area  $O(\delta)$ , there is a star-shaped set  $G$  with*

$$F \subset G \subset K, \quad G^\sharp = \{re^{i\theta} : 0 < r < 1 + \phi(\theta)\}, \quad \|\phi\|_{L^\infty} \leq \eta_0,$$

*such that*

$$|G \Delta E| \leq C\sqrt{\delta}, \quad \delta_T(G^\sharp) \leq C\delta. \quad (6.1)$$

Assuming Conjecture 6.1, the proof closes:

$$\mathcal{A}(G)^2 \leq C\delta_T(G^\sharp) \leq C\delta$$

by the direct radial-graph theorem, while

$$\mathcal{A}(E) \leq \mathcal{A}(G) + C|E \Delta G| \leq C\sqrt{\delta}.$$

Then  $\mathcal{A}(\Omega) \leq \mathcal{A}(E) + C\sqrt{\delta}$ , hence  $\mathcal{A}(\Omega)^2 \leq C\delta$ .

## 7 Attempted proof of the missing lemma

The natural proof is a decomposition of  $K \setminus F$ .

### 7.1 Closed holes

Closed holes are controlled by Lemma 3.2; their area is  $O(\delta)$ . Filling them changes asymmetry by  $O(\delta)$ . The torsion change is harmless either by the crude fill-rescale lemma or by the stronger capacity identity:

$$T(D) - T(D \setminus H) \geq a^2 \text{cap}_D(H)$$

whenever  $u_D \geq a$  on  $H$ . Tiny central holes are therefore not a real obstruction.

### 7.2 Boundary fjords

A fjord of radial depth  $h$  connected to the exterior has two sides of total length at least  $2h$ . Thus  $D_I \leq C\sqrt{\delta}$  excludes  $h \gg \sqrt{\delta}$ . Fjords of depth  $O(\sqrt{\delta})$  may be filled or ignored at the graph-amplitude level. If their area is  $O(\sqrt{\delta})$ , they sit in a torsion boundary layer of height  $O(\sqrt{\delta})$ , so their torsion effect should be  $O(\delta)$ . Making the last sentence rigorous requires a uniform boundary-layer comparison for the convex envelope  $G$ ; this is not written in the repo.

### 7.3 Outward fingers

For tube-like fingers the model estimate above proves the needed scale: surviving fingers on the  $\sqrt{\delta}$ -stopped level have length  $O(\delta^{3/4})$ , hence lie inside the graph amplitude threshold, and the ones not surviving have already been removed by the level flow. What remains missing is an arbitrary-geometry version:

$$\text{every non-graph outward component} \implies \text{a family of tube estimates like } D_H \simeq L/w. \quad (7.1)$$

This is not a cosmetic point. Without (7.1), the convex hull could overfill a sparse set of outward arcs, and the crude volume estimate would lose the sharp rate.

### 7.4 Bad rays

Plan 2's radial slicing controls multiplicity on bad rays once an annular bracket is known. At the  $\sqrt{\delta}$ -stopped level the annular bracket is available from the convex-hull/Bonnesen argument after deleting satellites. The remaining problem is not measuring the bad-ray set; it is proving that the radial repair along those bad rays changes torsion by  $O(\delta)$ , not merely that it changes area by  $O(\sqrt{\delta})$ .

## 8 Verdict

I cannot honestly claim the graph-reduction step is proved. But the  $\sqrt{\delta}$ -stopping scale makes the remaining theorem substantially more plausible and more precise.

What is now proved or reduced to standard planar facts:

1. a good level with

$$|\Omega \setminus E| \leq C\sqrt{\delta}, \quad D_H(E) + D_I(E) \leq C\sqrt{\delta}, \quad \delta_T(E) \leq C\delta;$$

2. satellite components have total area  $O(\delta)$ ;
3. closed holes with total perimeter  $O(\sqrt{\delta})$  have total area  $O(\delta)$ ;
4. the main connected component has a convex graph envelope with small  $L^\infty$  amplitude by planar convex-hull perimeter and Bonnesen;
5. model outward fingers are removed or made harmless at the correct scale by combining the level height  $t \leq C\sqrt{\delta}$  with the planar tube deficit and the  $D_H \simeq L/w$  estimate.

The exact missing point is Conjecture 6.1: an arbitrary-geometry decomposition showing that convexifying/radializing only the non-graph pieces of the good level changes the normalized torsion deficit by  $O(\delta)$ . This is narrower than the previous missing theorem and is where I would focus next.